MC Vertex Reweighting for PPG12:
A Data-Driven Iterative Method on the Truth Vertex

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Abstract

The PPG12 efficiency pipeline must reweight MC to match the measured $p+p$ luminous region, which is very different between the two crossing angles ($\sigma \simeq 53$ cm at 0 mrad vs $\sigma \simeq 22$ cm at 1.5 mrad, Piwinski narrowed). The production reco-level weight $f(z_r) = D(z_r)/M(z_r)$ closes the reco vertex distribution exactly, but it leaves the *individual* truth-vertex widths of double-interaction MC events at a geometric "half narrow, half wide" compromise ($\sigma \simeq 47$ cm instead of 22 cm), which in turn biases the truth fiducial cut used by the efficiency denominator. The resulting cross-section bias at 1.5 mrad is +1.4 % — comparable to other PPG12 systematics and well within reach of a correctable method. This note presents the nominal fix: an **iterative data-driven reweight** $w(z)$ applied per truth vertex ($w(z_h)$ for single interactions, $w(z_h)\,w(z_{\mathrm{mb}})$ for doubles). The method uses no physics prior for the target shape — only the empirical width of the data reco distribution as the seed Gaussian, and the 99-th percentile of $|z_r^{\mathrm{data}}|$ as the auto-derived closure window. Applied to the photon10 mix, the method converges cleanly at 0 mrad ($\max|D/M-1| < 1\,\%$ in 8 iterations) and reduces χ^2 in the closure window by a factor ~ 3100 at 1.5 mrad, closing the bulk of the window to the 1–2 % level inside $|z_r| < 30$ cm and reaching $\sim 11\,\%$ at the closure-window edge. The code is a standalone Python package in `efficiencytool/truth_vertex_reweight/`; the per-period `reweight.root` outputs are ready to be consumed by `RecoEffCalculator_TTreeReader.C`.

Contents

1 Problem and motivation

PPG12 extracts the isolated prompt-photon cross section from jet-dominated $p{+}p$ triggered data. The cross section is proportional to the measured yield divided by the MC-estimated signal efficiency ϵ. Both the numerator (reco clusters in data, passing the selection) and the denominator (MC truth photons in the fiducial window) inherit the MC vertex distribution — the MC prior on the vertex z-coordinate enters via (i) cluster kinematics reconstructed relative to the reco vertex z_r, (ii) the $|z_r| <$ `vertex_cut` cut applied in data and MC alike, and (iii) the truth fiducial cut $|z_h| <$ `vertex_cut` used for the efficiency denominator. *Any* data/MC mismatch in the vertex z distribution (reco or truth) therefore feeds directly into ϵ and the extracted cross section.

Notation. Throughout this note, z_r denotes the MBD-reconstructed vertex; $z_h \equiv z_{t,1}$ denotes the primary (hard-scatter) truth vertex; and $z_{\rm mb} \equiv z_{t,2}$ denotes the secondary (minimum-bias) truth vertex, present only in double-interaction events. For double-interaction MC, $z_r \approx (z_h + z_{\rm mb})/2$ up to MBD resolution. Sections 1–2 occasionally use the $(z_{t,1}, z_{t,2})$ pair notation when the algebraic decomposition into symmetric/antisymmetric coordinates is needed; elsewhere we use the $(z_h, z_{\rm mb})$ notation exclusively.

The two periods are very different. At the 0 mrad crossing angle the beams overlap along their full length and the luminous region is $\sigma_z \simeq 53$ cm. At 1.5 mrad the beams cross at an angle, the beam-beam overlap is geometrically shortened by the Piwinski factor, and the luminous region collapses to $\sigma_z \simeq 22$ cm. The MC samples used in PPG12 were generated with the full ~ 65 cm beam profile regardless of period; the reweight has to correct this downward by nearly a factor of three at 1.5 mrad. This is a very large shape correction and any shortcomings of the reweight method are amplified at 1.5 mrad.

The production method. The nominal PPG12 reweight (`efficiencytool/RecoEffCalculator_TTreeReade` lines 1331–1372) uses a 1D reco-vertex weight

$$f(z_r) \;=\; \frac{D(z_r)}{M_{\rm reco}(z_r)},$$

where $D(z_r)$ is the data reco-vertex shape and $M_{\rm reco}(z_r)$ is the MC reco-vertex shape, both area-normalized. Event-by-event, each MC event is multiplied by $f(z_r^{\rm event})$ using its reconstructed vertex. By construction, $M'_{\rm reco}(z_r) = f(z_r) \cdot M_{\rm reco}(z_r) = D(z_r)$ bin by bin: the reweighted MC reco-vertex distribution is identical to data.

Why the production method is not enough. $f(z_r)$ closes the reco distribution exactly, but it does not close the individual truth-vertex distributions of double-interaction MC events, and it is those truth vertices that enter the efficiency denominator via the truth fiducial cut. The structural reason is a clean algebraic one: decompose the pair $(z_{t,1}, z_{t,2}) = (z_h, z_{\rm mb})$ of truth vertices for a double-interaction event into a symmetric coordinate $\bar{z} \equiv (z_{t,1} + z_{t,2})/2$ and an antisymmetric coordinate $\varepsilon \equiv (z_{t,1} - z_{t,2})/2$. Under the independent Gaussian sampling of the raw MC, $\bar{z}$ and ε are statistically independent zero-mean Gaussians with $\sigma_{\bar{z}} = \sigma_\varepsilon = \sigma_{\rm MC}/\sqrt{2}$. The reco vertex is $z_r \approx \bar{z}$ (up to MBD resolution), so the production weight $f(z_r)$ depends *only* on $\bar{z}$ and is flat in ε. After reweighting, $\bar{z}$ matches the target Gaussian $\mathcal{N}(0, \sigma^2_{\rm lum}/2)$ by construction but ε is unchanged. The primary truth vertex $z_h = \bar{z} + \varepsilon$ is therefore a sum of a narrowed and an un-narrowed Gaussian,

$$\sigma^2(z_h) \;=\; \tfrac{1}{2}\,\sigma^2_{\rm lum} \;+\; \tfrac{1}{2}\,\sigma^2_{\rm MC} \;=\; \tfrac{1}{2}(22.3^2 + 62.7^2) \;\simeq\; (47.0\,{\rm cm})^2. \tag{1}$$

$f(z_r)$ cannot touch ε, so it cannot narrow z_h and $z_{\rm mb}$ individually below this "half-narrow, half-wide" compromise. Any principled fix must therefore act on the truth vertices directly, i.e. it must factorize as $w(z_h)\,w(z_{\rm mb})$ rather than depending only on the reco combination z_r. The nominal method of Section 3 does exactly this, with a data-driven $w(z)$ iterated to closure on the mixed-MC reco distribution.

2 Cross-section impact at 1.5 mrad

The efficiency denominator in `RecoEffCalculator_TTreeReader.C` uses both a reco-vertex cut $|z_r| <$ `vertex_cut` and a truth-vertex cut $|z_h| <$ `vertex_cut`, with `vertex_cut` $= 60\,\text{cm}$ in `config_bdt_nom.yaml`. The reco cut is handled correctly by $f(z_r)$ because the reco distribution is closed to data exactly. The truth cut is *not* handled correctly because, as Eq. (1) shows, $f(z_r)$ leaves $\sigma(z_h) \simeq 47\,\text{cm}$ instead of the target 22 cm for double-interaction MC.

Quantitative estimate. For a Gaussian truth distribution of width σ, the fraction in $|z_h| < 60\,\text{cm}$ is $\text{erf}(60/(\sigma\sqrt{2}))$. Plugging in:

- $\sigma = 47\,\text{cm}$ (what nominal $f(z_r)$ gives): $\text{erf}(60/(47\sqrt{2})) \simeq 0.80$
- $\sigma = 22\,\text{cm}$ (correct 1.5 mrad): $\text{erf}(60/(22\sqrt{2})) \simeq 0.99$

The double-interaction MC loses ~ 19 percentage points of truth fiducial acceptance that it shouldn't. The mixed MC blend at 1.5 mrad is 0.921single+0.079double, so the bias on the integrated efficiency denominator is

$$0.079 \times 0.19 \;\simeq\; 1.4\,\% \quad \text{at } 1.5\,\text{mrad}.$$

This translates directly to a +1.4 % bias on the extracted cross section at 1.5 mrad (the efficiency denominator is too small, so the quoted cross section is too large). At 0 mrad the analogous bias is $\sim 0.3\,\%$ because the MC beam profile is already close to the (wider) 0 mrad luminous region. At the percent level the current PPG12 production is biased at 1.5 mrad because $f(z_r)$ does not correct the individual truth vertices z_h and z_{mb} of double-interaction events. The bias is comparable to other PPG12 systematics and is easily fixable with a truth-level reweight.

3 Method: data-driven iterative reweight

The method derives a truth-vertex weight function $w(z)$ — evaluated at z_h for single-interaction MC and at both z_h and z_{mb} for double-interaction MC — directly from data, with no Gaussian assumption on the luminous-region shape, by iterating the mixed-MC reco distribution to closure against the data reco distribution.

Forward model. Let f_d be the cluster-weighted double-interaction fraction in a given crossing angle (0.079 at 1.5 mrad, 0.224 at 0 mrad) and $f_s = 1 - f_d$ the single-interaction fraction. The mixed MC reco distribution under a candidate truth-vertex weight $w(z)$ is

$$M(z_r) \;=\; f_s \sum_{e \in T_s} w(z_h^e)\, \delta(z_r - z_r^e) \;+\; f_d \sum_{e \in T_d} w(z_h^e)\, w(z_{\text{mb}}^e)\, \delta(z_r - z_r^e), \tag{2}$$

evaluated event-by-event over the single-interaction sample T_s and the double-interaction sample T_d. For double-interaction events the weight enters *twice*, once for the primary truth vertex z_h and once for the secondary truth vertex z_{mb}, reflecting the statistical independence of the two truth vertices under the generator. This is the crucial structural choice: it gives the iteration access to the antisymmetric coordinate ε of Section 1 that the production reco-only weight $f(z_r)$ cannot touch.

Binning and closure window. Both periods use the same binning: 80 bins over $[-200, 200]\,\text{cm}$ (bin width 5 cm), no overflow clipping, and identical bin edges for the $w(z)$ weight function and the reco-vertex histograms. The iterative closure window is auto-derived per period as the 99-th percentile of $|z_r^{\text{data}}|$:

- 1.5 mrad: $z_{\text{cut}} = 83\,\text{cm}$ (34 bins inside), from $|z_r^{\text{data}}|_{99\%} = 82.0\,\text{cm}$.
- 0 mrad: $z_{\text{cut}} = 144\,\text{cm}$ (58 bins inside), from $|z_r^{\text{data}}|_{99\%} = 143.9\,\text{cm}$.

The window is purely data-driven and requires no manual tuning. Outside it, the statistics in data drop below the level where the iteration could meaningfully constrain $w(z)$.

Seed. The iteration starts from a Gaussian seed $w_0(z) = \mathcal{N}(0, \sigma^2_{\text{seed}}) \,/\, \mathcal{N}(0, \sigma^2_{\text{gen}})$, with σ_{gen} measured from the photon10 primary truth vertex z_h (`vertexz_truth` branch) ($\sigma_{\text{gen}} = 64.997 \pm 0.015$ cm, intrinsic to the generator and identical between periods, since the same single MC is loaded for both) and $\sigma_{\text{seed}} = \text{std}(z_r^{\text{data}})$ measured directly from the data reco-vertex distribution in the relevant run range (21.48 cm at 1.5 mrad, 54.32 cm at 0 mrad). Note that σ_{seed} requires *no* physics input — no Piwinski narrowing factor, no luminous-region calculation. It is just the empirical width of the reco distribution we are trying to match.

Iterative update. At iteration i, the residual is $r_i(z_r) = D(z_r)/M_i(z_r)$, where $D(z_r)$ is the data reco-vertex histogram and $M_i(z_r)$ is the mixed MC reco-vertex histogram from Eq. (2) under the current w_i. The update is Lucy–Richardson-like, multiplicative, and damped:

$$w_{i+1}(z) \;=\; w_i(z) \cdot \big[\text{smooth}(r_i(z))\big]^{\alpha}, \qquad \alpha = 0.5, \tag{3}$$

where the residual r_i evaluated at reco bin z_r is applied to the weight at the matching truth bin $z = z_r$ (using the shared binning). After each update, w is renormalized so that $\langle w(z_h) \rangle$ averaged over the single-MC truth distribution equals unity, preserving the physics double-interaction fraction. Damping ($\alpha = 0.5$) prevents over-correction; a 1-bin Gaussian smoothing on the residual prevents bin-by-bin oscillation; the update factor is clipped to $[0.5, 2]$ per iteration to bound jumps; and the endpoints of the update region are edge-clamped to their nearest interior values to prevent runaway behavior at the boundary. The update is applied over the full 80-bin binning and the final $w(z)$ is data-driven everywhere, with no post-iteration snap to a seed Gaussian. (An earlier version of the method snapped $w(z)$ to the seed Gaussian outside the closure window; a systematic study with `compare_snap_strategies.py` showed that removing the snap improves 0 mrad closure by a factor ~ 1.3 in χ^2 and ~ 3 in $\max |r - 1|$ while leaving 1.5 mrad essentially unchanged.) Convergence is tested as $\max |r_i(z) - 1| < 0.01$ inside the closure window, with a hard cap of 30 iterations.

Implementation. The method is a standalone Python tool in `efficiencytool/truth_vertex_reweight/`. The mixed-MC histogram is heavy to rebuild (~ 22 M events at 1.5 mrad, ~ 48 M at 0 mrad), so per-sample arrays are cached to `output/cache/*.npz` keyed on file mtime; each iteration runs in seconds per iteration on a single core. A short file list is in Appendix D. Per-period outputs live in `output/{1p5mrad,0mrad}/` and include `reweight.root` (histograms `h_w_iterative`, `h_w_seed`, `h_M_iterative`) and `fit_result.yaml`.

4 Per-period results

Both periods are fit on the photon10 / photon10_double mix only at this stage; the full 4-sample blend including jet12/jet12_double is a follow-up (Section 6). Table 1 collects the headline numbers from both fits.

quantity	1.5 mrad	0 mrad
run range	51274–54000	47289–51274
luminosity [pb^{-1}]	16.8588	32.6574
f_{double} (cluster-weighted)	0.079	0.224
data $\lvert z_r\rvert$ 99% percentile [cm]	82.0	143.9
closure window z_{cut} [cm]	83	144
bins inside closure window	34	58
$\sigma_{\mathrm{seed}} = \mathrm{std}(z_r^{\mathrm{data}})$ [cm]	21.48	54.32
σ_{gen} from MC z_h (generator) [cm]	64.997 ± 0.015	
initial χ^2 (in window)	1,419,454	26,589
final χ^2 (in window)	453	158
χ^2 reduction factor	3132×	168×
final χ^2/ndf (in window)	13.7 (ndf 33)	2.78 (ndf 57)
initial max $\lvert r-1\rvert$	0.635	0.079
final max $\lvert r-1\rvert$ (in window)	0.111	0.008
iterations used	30 (cap hit)	8
converged to 0.01 tolerance	**no**	**yes**
(context) parametric Gaussian χ^2/ndf	3237	454

Table 1: Iterative data-driven reweight, photon10 mix only, both crossing angles, with the post-iteration snap to seed removed (Section 3). The 0 mrad fit converges to the 1 % per-bin tolerance in 8 iterations (final max $\lvert r-1\rvert \simeq 0.8\,\%$). The 1.5 mrad fit reduces χ^2 in the closure window by a factor ~ 3100 in 30 iterations and stalls at max $\lvert r-1\rvert \simeq 11\,\%$, dominated by bins at the closure-window edge $\lvert z_r\rvert \simeq 83$ cm. Inside the bulk $\lvert z_r\rvert < 30$ cm the per-bin closure is at the 1–2 % level (see Figure 4). The last row reports the parametric Gaussian χ^2/ndf (best-fit σ_{tgt} from Stage-1) against the data; the iterative method improves on this by a factor of ~ 236 at 1.5 mrad and ~ 163 at 0 mrad.

Hero summary. Figure 1 is the single-figure summary: data reco vs reweighted MC reco overlaid for both periods, residuals, the recovered $w(z)$, and the headline numbers. The 0 mrad mix-MC matches data to $\lesssim 1\,\%$ per bin throughout the closure window; the 1.5 mrad mix matches to ~ 1–$2\,\%$ in the bulk $\lvert z_r\rvert < 30$ cm, degrades to ~ 6–$8\,\%$ around $\lvert z_r\rvert \simeq 60$ cm, and reaches $\sim 11\,\%$ at the closure-window edge $\lvert z_r\rvert \simeq 83$ cm.

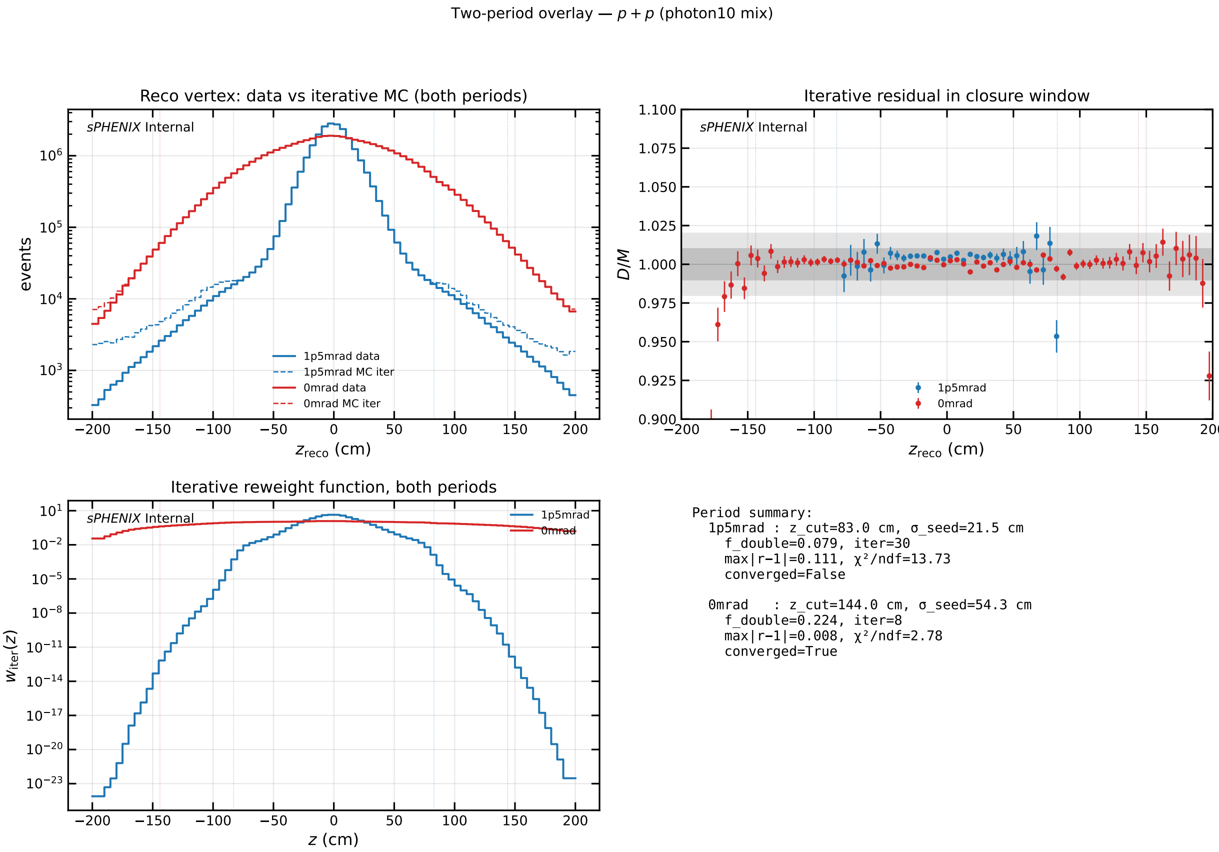

Figure 1: Iterative data-driven reweight summary, both crossing angles (2×2 grid plus summary text). Top-left: data reco z_r (solid) vs mixed MC reco z_r after iterative reweight (dashed), both periods overlaid (log scale). Top-right: residual $r(z_r) = D(z_r)/M_{\mathrm{iter}}(z_r)$ inside each period's closure window (vertical dashed lines). Bottom-left: recovered truth-vertex weight $w(z)$ for both periods (log scale). Bottom-right: summary statistics. The 0 mrad case is essentially closed; the 1.5 mrad case is closed in the bulk and stalls at the closure-window edge (see Section 5).

Truth vertex distribution comparison. Figure 2 overlays the primary truth vertex z_h distribution for the mixed MC under three weighting methods: no reweight (raw MC, $\sigma \simeq 64$ cm), the production reco-level weight $f(z_r)$, and the new iterative truth-level weight $w(z_h)$. At 1.5 mrad the production method leaves the mixed-MC primary truth vertex z_h at $\sigma \simeq 27$ cm — narrower than the raw MC but still far wider than the true luminous-region width ($\sigma_{\mathrm{lum}} \simeq 22$ cm), because $f(z_r)$ cannot narrow z_h and z_{mb} individually for double-interaction events below the half-narrow/half-wide compromise (Section 1). The iterative method $w(z_h)$ narrows the z_h distribution to $\sigma \simeq 16$ cm, which is narrower than the data reco width $\sigma_{\mathrm{seed}} = 21.5$ cm precisely as expected, since each single-interaction truth vertex must be narrower than the observed reco width (the MBD vertex finder adds resolution smearing). At 0 mrad, where the correction is small, both methods converge to the same $\sigma \simeq 54$–57 cm.

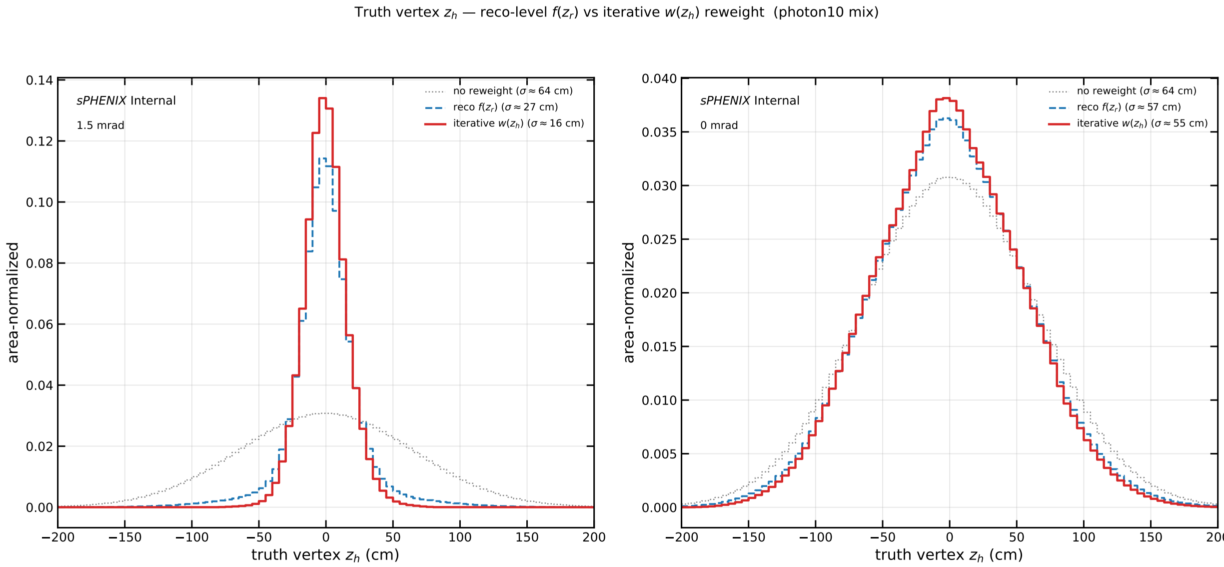

Figure 2: Truth vertex z_h distribution for the mixed photon10 MC under three reweighting methods, per period. Left: 1.5 mrad, where the reco-level $f(z_r)$ (blue dashed) fails to narrow the truth distribution sufficiently, while the iterative $w(z_h)$ (red) narrows it well past the data reco width (the truth must be narrower than the reco due to MBD resolution smearing). Right: 0 mrad, where both methods agree because the MC beam profile is already close to the data luminous region.

0 mrad: clean convergence. Figure 3 shows the 6-panel diagnostic for 0 mrad. The seed Gaussian with $\sigma_{\text{seed}} \simeq 54\,\text{cm}$ already captures most of the shape (initial $\max|r-1| = 8\,\%$, initial $\chi^2/\text{ndf} \simeq 467$); the parametric best-fit Gaussian at $\sigma_{\text{tgt}} \simeq 54.6\,\text{cm}$ is essentially the same ($\chi^2/\text{ndf} \simeq 454$) because the data is nearly Gaussian. The iteration cleans up the residual over 8 updates and drops below the 1 % per-bin tolerance inside the closure window (final $\max|r-1| \simeq 0.8\,\%$, $\chi^2/\text{ndf} \simeq 2.78$). The recovered $w(z)$ is smooth and broadly Gaussian, with a small enhancement of the central core relative to the seed, consistent with the data luminous region being slightly more peaked than a single Gaussian.

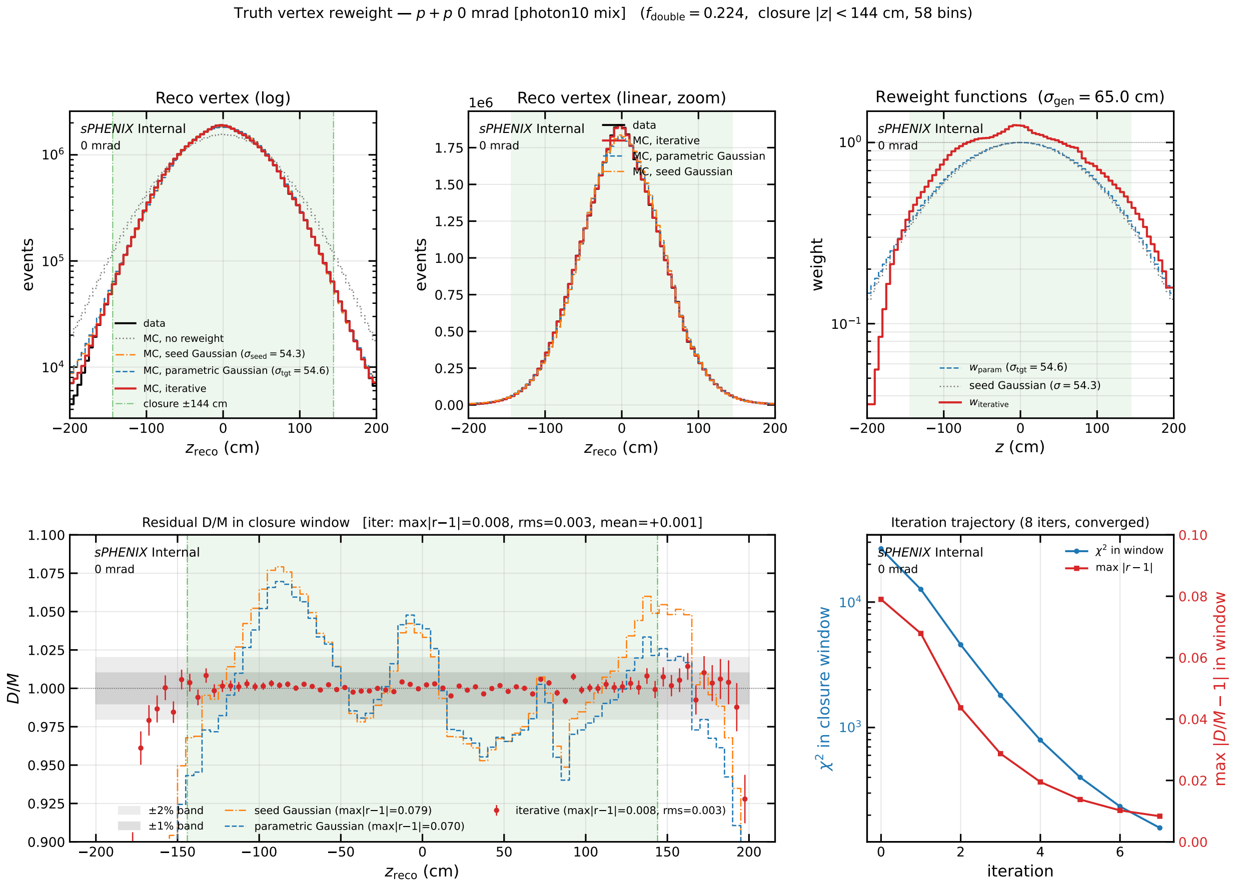

Figure 3: 0 mrad iterative closure (6 panels). Top-left/centre: data reco vs mixed MC reco under no reweight (gray dotted), the seed Gaussian $w(z) \propto \mathcal{N}(0, \sigma^2_{\text{seed}})/\mathcal{N}(0, \sigma^2_{\text{gen}})$ with $\sigma_{\text{seed}} = \text{std}(z_r^{\text{data}})$ (orange dash-dot), the parametric best-fit Gaussian with σ_{tgt} from the χ^2 scan (blue dashed), and the iterative reweight (red). Top-right: the corresponding reweight functions $w(z)$ (seed, parametric, iterative). Bottom-left: residual D/M in the closure window with Poisson error bars on the iterative points and $\pm 1\,\%/\pm 2\,\%$ reference bands. Bottom-right: per-iteration χ^2 and max $|r-1|$ trajectory. At 0 mrad the seed and parametric Gaussians are nearly identical ($\sigma_{\text{seed}} \simeq \sigma_{\text{tgt}} \simeq 54\,\text{cm}$) because the data reco distribution is already close to a single Gaussian; the iterative method improves on both.

1.5 mrad: bulk closure, edge stall. Figure 4 shows the 6-panel diagnostic for 1.5 mrad. The story is qualitatively different. The seed Gaussian with $\sigma_{\text{seed}} \simeq 21\,\text{cm}$ starts from max $|r-1| = 64\,\%$ and an in-window χ^2 of 1.42×10^6; the first ~ 5 iterations make dramatic progress, then the iteration enters a slow plateau driven by bins at the closure-window edge $|z_r| \simeq 83\,\text{cm}$. After 30 iterations, χ^2 in the window has dropped by $\sim 3100\times$ to $\simeq 453$, and max $|r-1|$ sits at $\simeq 0.11$, set by those stuck edge bins. Inside the bulk $|z_r| < 30\,\text{cm}$ closure is at the ~ 1–$2\,\%$ level on every bin; between 30 and 60 cm the residual climbs to ~ 5–$10\,\%$. The recovered $w(z)$ shows a strong central core boost and a clear non-Gaussian shape in the $|z| \simeq 40$–$60\,\text{cm}$ region (truth-vertex axis) that no symmetric Gaussian ansatz can represent.

Truth vertex reweight — $p+p$ 1.5 mrad [photon10 mix] ($f_{\rm double} = 0.079$, closure $|z| < 83$ cm, 34 bins)

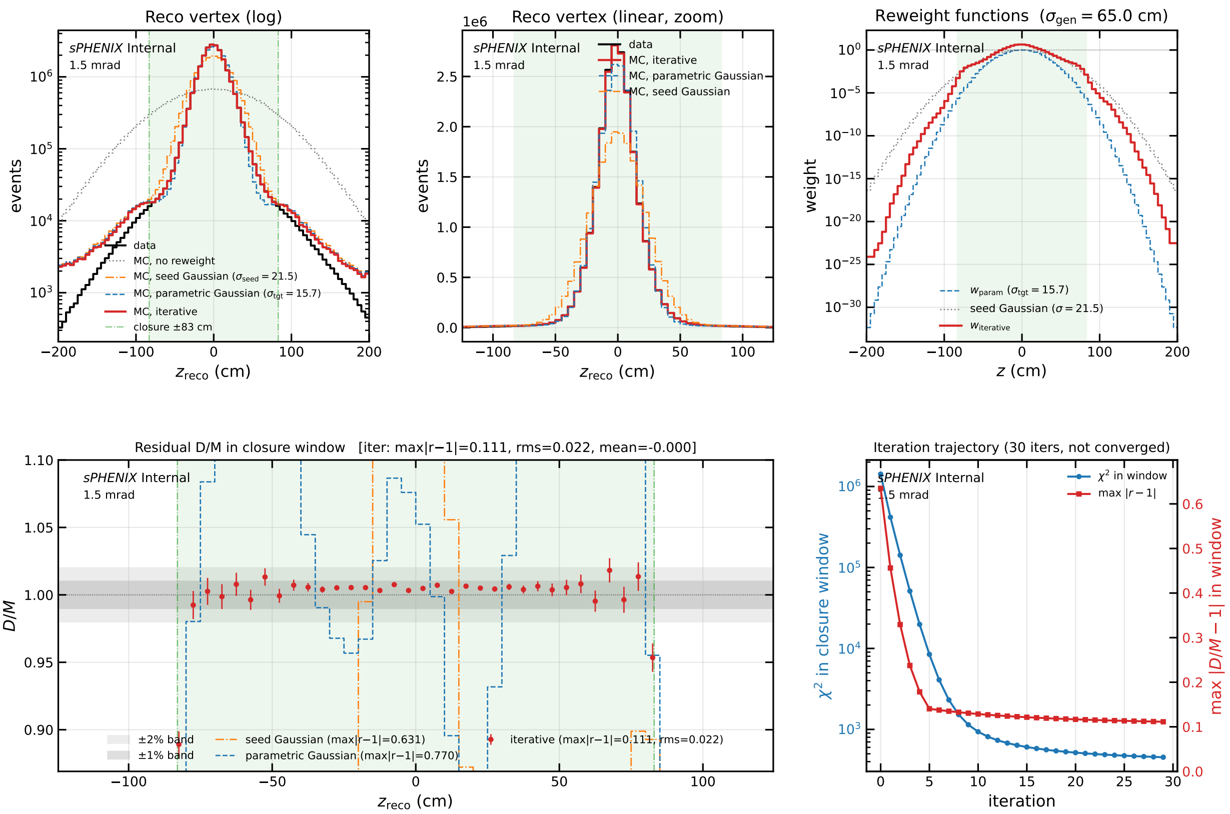

Figure 4: 1.5 mrad iterative closure (6 panels), same layout as Figure 3. At 1.5 mrad the seed Gaussian ($\sigma_{\rm seed} = 21.5$ cm, orange dash-dot) and the parametric best-fit Gaussian ($\sigma_{\rm tgt} = 15.7$ cm, blue dashed) are distinct; both are much broader than the data core and mis-predict the reco-vertex shape by $\sim$30–60 % at the closure-window edge. The iterative method (red points) closes the bulk of the window to $\sim$1–2 % inside $|z_r| < 30$ cm and stalls at max $|r-1| \simeq 0.11$ at the closure-window edge $|z_r| \simeq 83$ cm. The recovered $w(z)$ in the top-right panel shows a strong central core boost and a clear non-Gaussian shape in the $|z| \simeq 40$–60 cm region (truth-vertex axis).

Method comparison. Figure 5 puts the iterative reweight against three reference baselines at 1.5 mrad: no reweight, the seed Gaussian (the $\mathcal{N}(0, \sigma^2_{\rm seed})$ initial point of the iteration, with $\sigma_{\rm seed} = 21.5$ cm measured from std($z_r^{\rm data}$)), and the parametric best-fit Gaussian (a single-parameter fit scanning $\sigma_{\rm tgt}$ against data, best at $\sigma_{\rm tgt} = 15.7$ cm). (Stage 1+2, a parametric residual correction on top of the parametric Gaussian, is never used in the nominal pipeline and is not shown.) The no-reweight MC overshoots data by a factor of several everywhere; both Gaussian baselines close the width on average but mismatch the shape at the 30–70 % level, with visible structure in the ratio panel that no single-Gaussian ansatz can represent; the iterative method is the only one that follows the narrow data core while also tracking the non-Gaussian mid-tail structure.

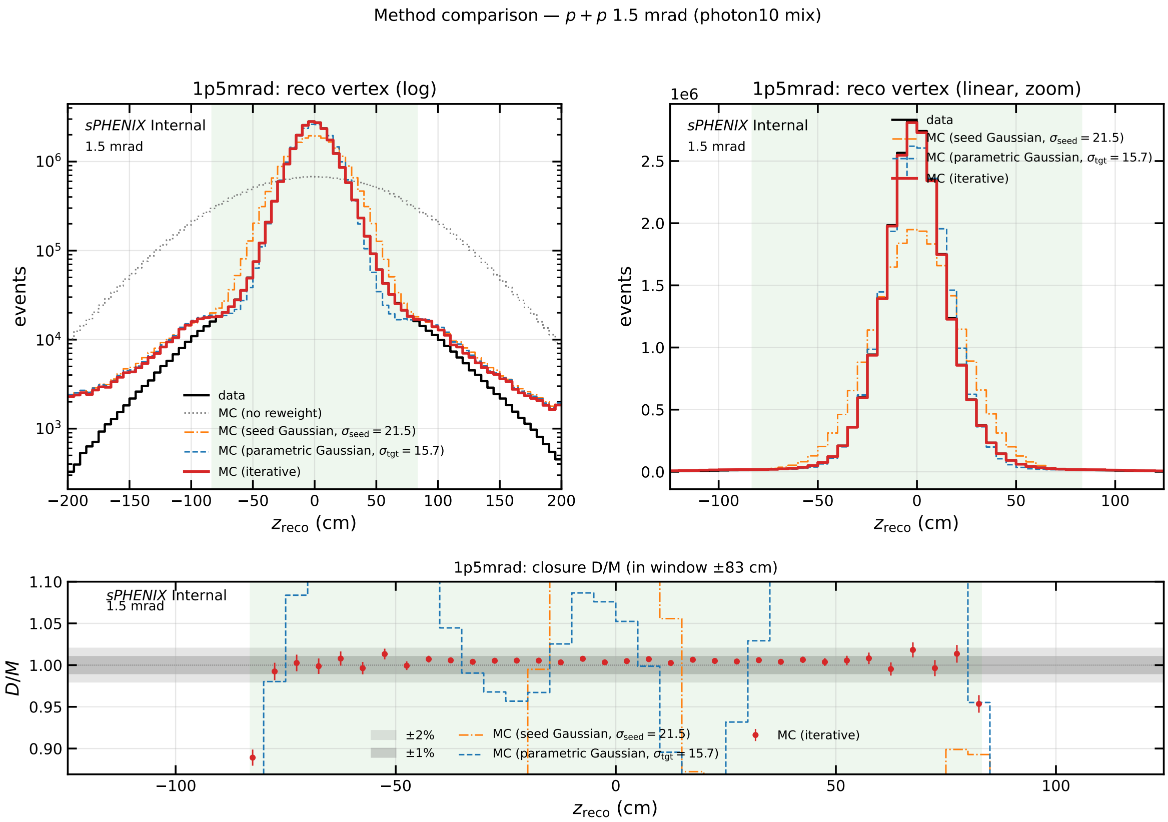

Figure 5: Method comparison at 1.5 mrad: data reco vs mixed MC reco under no reweight (gray dotted), seed Gaussian $\sigma_{\rm seed} = 21.5$ cm (orange dash-dot), parametric best-fit Gaussian $\sigma_{\rm tgt} = 15.7$ cm (blue dashed), and the iterative data-driven reweight (red), with a ratio panel below. The two Gaussian baselines bracket the "best possible" single-Gaussian description of the data reco shape; neither closes below the 10–30 % level because the true data-MC ratio at 1.5 mrad is non-Gaussian. The iterative method is the only one that follows the narrow data core while also tracking the non-Gaussian mid-tail structure.

Parametric $\sigma_{\rm tgt}$ scan. Figure 6 shows the stage 1 parametric χ^2 scan that precedes the iterative method. A single Gaussian truth-vertex weight $w(z) \propto \mathcal{N}(0, \sigma^2_{\rm tgt})/\mathcal{N}(0, \sigma^2_{\rm gen})$ is evaluated at each scan point. At 1.5 mrad the minimum is at $\sigma_{\rm tgt} \simeq 15.7$ cm with $\chi^2/{\rm ndf} \simeq 3237$; the iterative method reaches 13.7, a $\sim$236× improvement. At 0 mrad the minimum is at $\sigma_{\rm tgt} \simeq 54.6$ cm with $\chi^2/{\rm ndf} \simeq 454$; the iterative method reaches 2.78, a $\sim$163× improvement. The gap between the best parametric point and the iterative floor quantifies the shape mismatch that no single-Gaussian ansatz can capture.

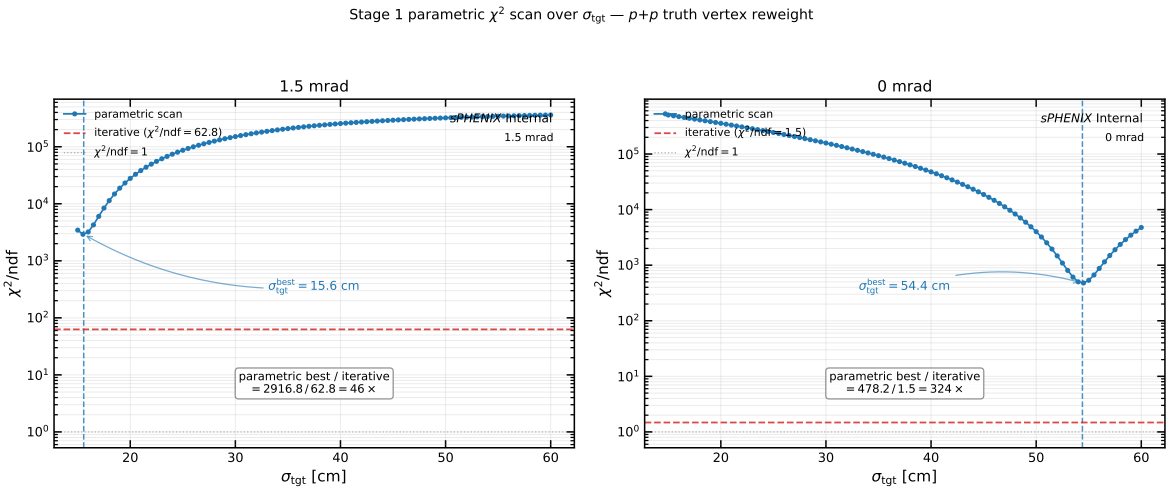

Figure 6: Stage 1 parametric χ^2/ndf as a function of the target truth-vertex Gaussian width σ_{tgt}, for the 1.5 mrad (left) and 0 mrad (right) periods. The minimum of the scan identifies the best single-Gaussian approximation to the data reco-vertex shape; the parabolic-refined best-fit σ_{tgt} is marked by a vertical dashed line. The horizontal dashed line (red) shows the final χ^2/ndf achieved by the iterative data-driven method, and the dotted line marks $\chi^2/\text{ndf} = 1$ for reference. All χ^2 values are computed on the reco-vertex axis z_r inside the closure window and normalized by the number of degrees of freedom.

5 Physics interpretation

Why 0 mrad is easy. At 0 mrad the beams overlap head-on along their full length and the luminous region is broad ($\sigma_{\text{lum}} \simeq 54$ cm), nearly equal to the generator beam profile ($\sigma_{\text{gen}} \simeq 65$ cm). The shape of the data reco vertex is also close to a single Gaussian. The seed is therefore already a good first approximation, the initial residual is ~ 8 %, and the iteration just cleans up small departures (slightly enhanced central core). Convergence in a handful of iterations is what one would expect from a near-textbook deconvolution problem.

Why 1.5 mrad is hard. Three things conspire at 1.5 mrad:

- The Piwinski crossing-angle effect compresses the luminous region to $\sigma_{\text{lum}} \simeq 21$ cm, almost a factor of three narrower than the generator profile. The reweight must extract a narrow target from a wide source, and the dynamic range of $w(z)$ at the central core is correspondingly large ($w(0) \simeq 8$).
- The data reco distribution is *not* a clean Gaussian. It has a narrow core plus a mid-tail shoulder near $|z_r| \simeq 45$–60 cm. The iterative method discovers this structure directly — the recovered $w(z)$ is non-Gaussian in the same z range — whereas any symmetric one-parameter Gaussian ansatz cannot represent it. This is why the parametric best-fit Gaussian gives $\chi^2/\text{ndf} \simeq 3237$ on the 1.5 mrad mix, and why the iterative method drops it to $\simeq 14$ (a further $\sim 236\times$ improvement).
- The MBD resolution tail leak from core-truth events fills the bins near the closure-window edge ($|z_r| \simeq 83$ cm). The iteration becomes insensitive to $w(z)$ updates at that z because the contributions to those reco bins come dominantly from truth vertices (z_h, z_{mb}) at other z values via the wide double-interaction reco-truth kernel. The iteration has the wrong handle on those bins, and that is what sets the residual floor at $\max|r - 1| \simeq 11$ %.

What the recovered $w(z)$ looks like. At 1.5 mrad, $w(z)$ has three distinguishing features visible in the top-right panel of Figure 4: (a) a strong central peak with $w(0) \simeq 8$ that pulls double-interaction events whose truth vertices (z_h and $z_{\rm mb}$) both lie close to the IP into the data-like core; (b) a mid-z dip that suppresses the generator's natural $|z| \sim 30$–50 cm population to match the narrow data core; and (c) a clear $\pm$ asymmetry in $|z| \simeq 40$–60 cm tracking the asymmetric mid-tail bump in the data. None of these features is representable by a symmetric single-parameter Gaussian. At 0 mrad, $w(z)$ is essentially the seed Gaussian with a small central enhancement.

6 Caveats and next steps

(i) photon10-only mix. The current fit uses `photon10 + photon10_double` only. The full production blend also includes `jet12 + jet12_double`. Extending the fit to all four samples is the obvious next step: mechanically it is two more `load_sample` calls plus the appropriate cluster-weighted blend, but the closure numbers will move and should be remeasured before any production hookup.

(ii) 1.5 mrad is not formally converged. The 30-iteration cap was hit. Two questions remain open: (a) whether extending to ~ 100 iterations would push the stuck edge bin to convergence, or whether it is a hard physical floor set by the double-interaction reco-truth kernel (the suspicion is the latter); and (b) whether the failure to converge in $\max|r-1|$ matters for the downstream analysis. The latter has a fairly clean answer: the PPG12 analysis applies fiducial cuts $|z_r| < 60$ cm on the reco vertex and $|z_h| < 60$ cm on the primary truth vertex (Section 2), and inside those cuts the per-bin closure is already at the 1–2 % level. The stuck edge bin at $|z_r| \simeq 83$ cm is well outside the analysis fiducial, so the iteration is good enough for the analysis use case, even though it is not formally converged on the full closure-window metric.

(iii) Downstream hookup is wired in but off by default. Both `ShowerShapeCheck.C` and `RecoEffCalculator_TTreeReader.C` now accept two new analysis fields, `truth_vertex_reweight_on` and `truth_vertex_reweight_file`; when `on=1`, the code opens `reweight.root`, reads `h_w_iterative`, and multiplies the per-event weight by $w(z_h)$ for single-interaction MC and $w(z_h)\,w(z_{\rm mb})$ for double-interaction MC. The secondary truth vertex branch `vertexz_truth_mb` (i.e. $z_{\rm mb}$) is picked up through an optional-pointer guard, so single-MC trees without it fall through to the single-vertex path. The two reco-level and truth-level reweights are mutually exclusive and the config loader enforces this by forcing `vertex_reweight_on=0` when `truth_vertex_reweight_on=1`. The showershape configs `config_showershape_{1p5rad,0rad}.yaml` are pre-wired to the matching `output/{1p5mrad,0mrad}/reweight.root`; the nominal BDT configs carry the same fields disabled, so flipping the main analysis over to the truth-vertex reweight is a one-line config edit once the jet-sample extension is done.

(iv) Implementation can be sped up. Each iteration currently rebuilds the MC reco histogram from scratch ($\mathcal{O}(N_{\rm events})$ per pass). Precomputing z^2 and dropping the per-trial mask rebuild would give a $\sim 10\times$ speedup with no change in result. Not a blocker; nice-to-have for the all-samples extension.

7 Summary

$f(z_r)$ cannot close the individual truth-vertex distributions (z_h and $z_{\rm mb}$) of double-interaction MC, which biases the primary truth fiducial cut $|z_h| < 60$ cm used in the efficiency denominator by $+1.4$ % at 1.5 mrad (Section 2). An iterative data-driven reweight $w(z)$, applied per truth vertex with the factorization $w(z_h)\,w(z_{\rm mb})$ for double-interaction events, closes this gap without any Gaussian assumption. Applied to the photon10 mix, the method converges cleanly at 0 mrad ($\max|r-1| < 1$ % in 8 iterations) and reduces in-window χ^2 by a factor ~ 3100 at 1.5 mrad, closing the analysis-relevant bulk $|z_r| < 30$ cm to 1–2 % per bin and reaching ~ 11 % at the closure-window edge. The recovered $w(z)$ at 1.5 mrad shows non-Gaussian features (a strong central boost and a mid-z shoulder) that no closed-form Gaussian ansatz could represent. Per-period `reweight.root` outputs are ready in `efficiencytool/truth_vertex_reweight/output/` and the C++ hookup into `ShowerShapeCheck.C` and `RecoEffCalculator_TTreeReader.C` is in place (enabled on the showershape period configs, wired but off on the nominal BDT configs). The remaining work is the jet-sample extension of the standalone fit tool.

A Baseline: events with and without a reconstructed vertex

Before building any truth-based reweight, we checked that dropping events whose reco MBD vertex is the -9999 sentinel does not itself bias the MC. The concern is that an MC event which loses its reco vertex z_r may be physically unusual (e.g. with a shifted truth vertex z_h), so dropping it would skew the efficiency in a z_h-dependent way.

Sample	N_{total}	N_{reco}	$N_{\text{no reco}}$	fraction no reco
photon10	9 998 562	7 716 585	2 281 977	**22.82 %**
photon10_double	9 997 075	9 491 223	505 852	**5.06 %**
jet12	10 000 000	7 689 530	2 310 470	**23.10 %**
jet12_double	9 997 000	9 485 384	511 616	**5.12 %**

Table 2: Fraction of events without an MBD reco vertex. The double-interaction productions have $\sim 4.5\times$ higher MBD vertex-finding efficiency than their single-interaction counterparts (two concurrent collisions give the MBD twice the chance of producing a vertex).

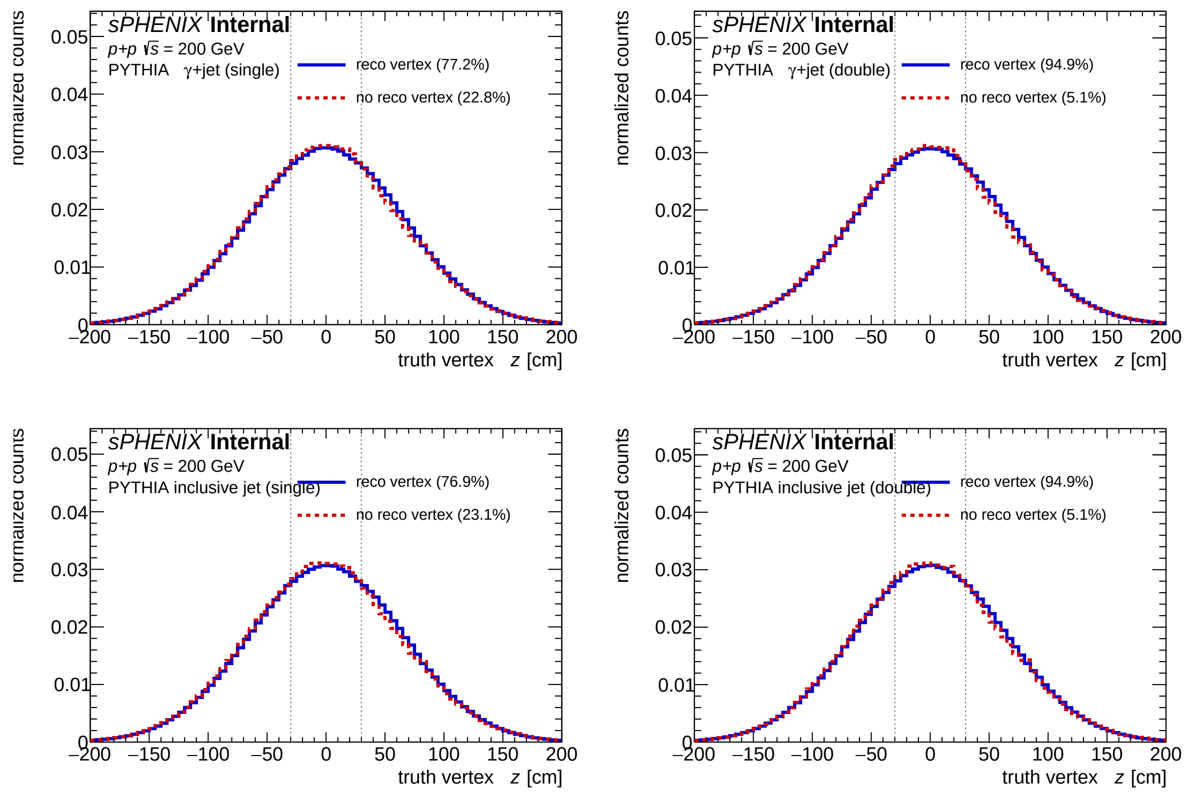

Figure 7: Primary truth vertex z_h distribution for events with (blue) and without (red dashed) a reconstructed MBD vertex z_r, area-normalized for shape comparison. Columns: single / double interaction. Rows: photon10 / jet12. Dotted vertical lines: $|z_h| = 30$ cm reference.

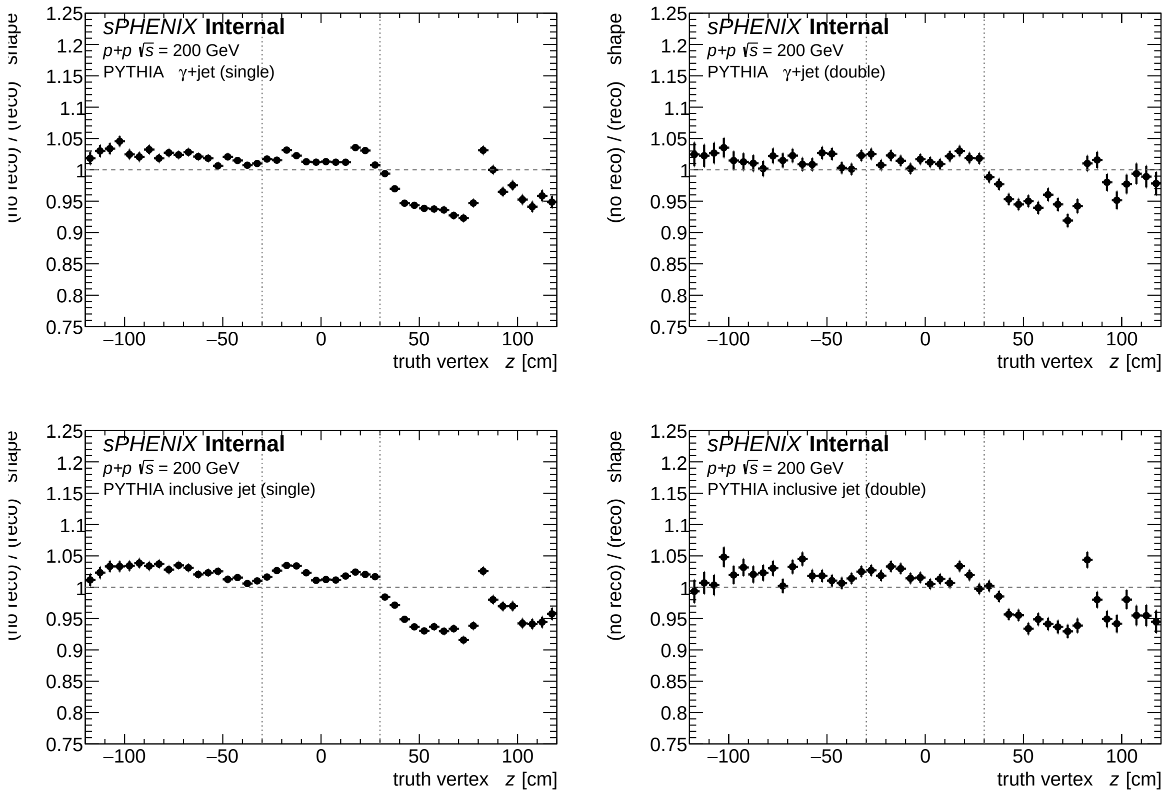

Figure 8: Ratio (no-reco-vertex shape) / (reco-vertex shape) as a function of primary truth vertex z_h, zoomed to $[0.75, 1.25]$. All samples show a $\sim\pm5\,\%$ north/south asymmetry inside $|z_h| < 30\,\mathrm{cm}$, consistent with a small MBD geometric bias.

Inside $|z_h| < 30\,\mathrm{cm}$ the two shapes agree to within $\pm5\,\%$ with a small z_h-asymmetry that is the same in all four samples and therefore a detector-level effect (MBD vertex finder slightly more efficient at positive z). Dropping the no-reco-vertex events is thus *approximately* harmless — both the efficiency numerator and denominator are scaled by the same factor, and the residual z dependence of the drop rate is small. However, dropping $\sim22\,\%$ of the single-interaction MC sample is still a lot of statistics, and it makes the efficiency measurement implicitly conditional on a valid reco vertex being found. That was the original motivation for exploring a truth-level reweight in the first place.

B Discarded methods

Before arriving at the iterative method described in Section 3, we explored three truth-only reweight constructions that did not work:

- w_1: **direct ratio** of the data reco-vertex shape to the MC truth-vertex shape, $w_1(z_h) = D(z_r)/T(z_h)$, with the common binning identifying z_r bins with z_h bins.
- w_2: **conditional projection**, $w_2(z_h) = \mathbb{E}[f(z_r)\,|\,z_h]$, the expectation of the production reco weight conditional on the primary truth vertex.
- $w_{\mathrm{iter,DA}}$: **D'Agostini iterative unfolding** using the MC response matrix $R_{ij} = P(z_r^i|z_h^j)$.

All three hit a geometric lower bound on the double-MC sample at 1.5 mrad: the conditional reco kernel $P(z_r|z_h)$ (for single-interaction MC) has RMS $\sim34\,\mathrm{cm}$, wider than the $\sim22\,\mathrm{cm}$ target data peak (Section C), so *any* truth-only reweight is bounded from below by the kernel width and cannot reach the target.

We also drafted a closed-form **2D Gaussian reweight** $w(z_h, z_{\rm mb}) \propto \exp[-\frac{1}{2}(z_h^2 + z_{\rm mb}^2)\,\Delta]$ with $\Delta = 1/\sigma_{\rm lum}^2 - 1/\sigma_{\rm MC}^2$. This closed form does work in principle — it acts on both the symmetric and antisymmetric coordinates and so can narrow all three vertex axes simultaneously — but it assumes a single Gaussian shape for the data luminous region. At 1.5 mrad the data has a $\pm$-asymmetric mid-tail bump that a symmetric Gaussian cannot reproduce, so the closed form ended up as a first-draft proposal that was superseded by the iterative method of Section 3, which discovers $w(z)$ from data directly without any Gaussian assumption.

C Double-interaction reco-truth kernel

The geometric lower bound on truth-only reweights quoted in Section B can be read off the widths directly:

- 1.5 mrad data reco width: $\sigma \simeq 22.3$ cm
- jet12 single-interaction MC truth width (z_h): $\sigma \simeq 57.6$ cm
- jet12_double MC truth width (z_h, primary): $\sigma \simeq 62.7$ cm
- jet12_double MC reco width (z_r): $\sigma \simeq 48.6 \approx 62.7/\sqrt{2}$ (averaging)
- Conditional kernel $P(z_r|z_h{=}0)$ RMS (with $z_{\rm mb}$ marginalized): ~ 34 cm

Figure 9 shows the kernel slice directly, compared to the 1.5 mrad data target. Even if the primary truth vertex is forced to $z_h = 0$, the reco distribution from a double-interaction event cannot be narrower than the ~ 34 cm kernel width — larger than the 22 cm target. This is the structural reason the truth-only methods fail on double-interaction MC, and the structural reason the iterative method needs the $w(z_h)\,w(z_{\rm mb})$ factorization rather than a w-on-z_r construction.

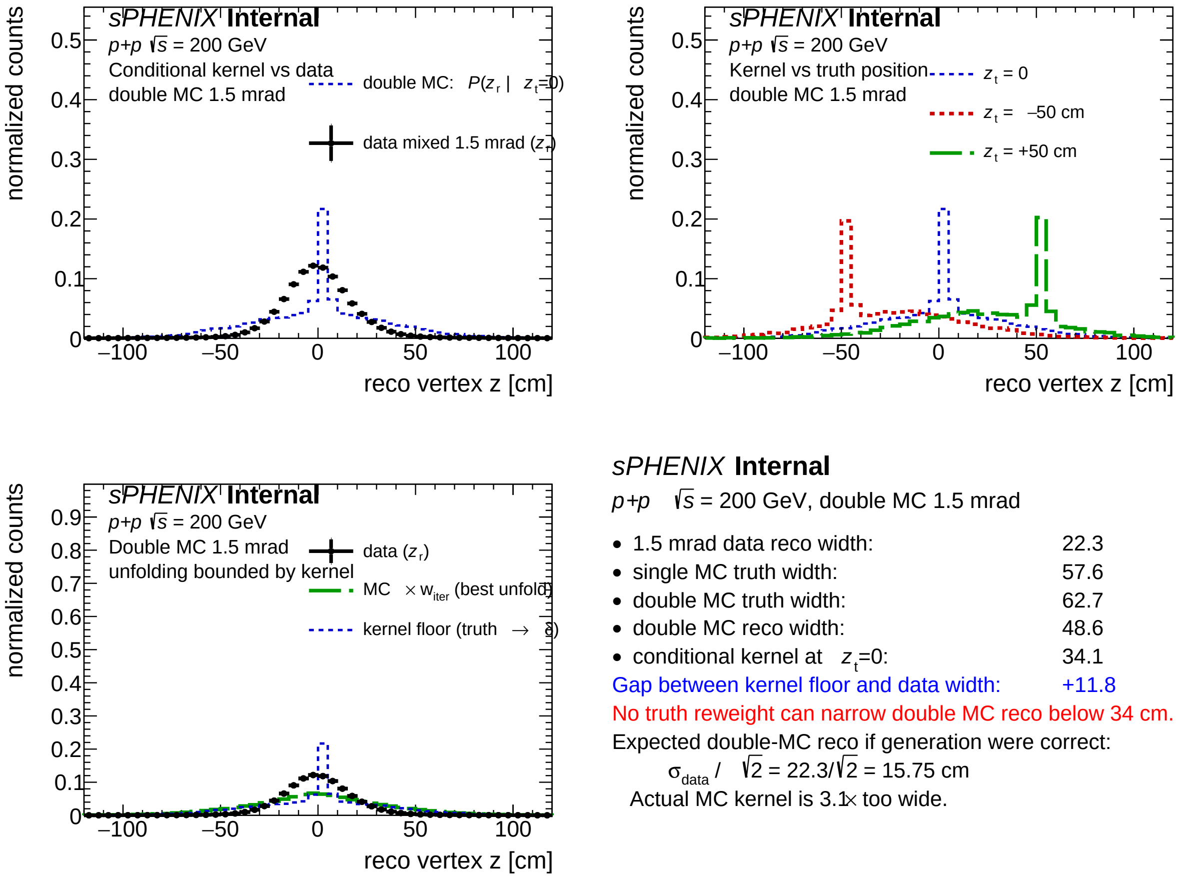

Figure 9: Conditional reco kernel $P(z_r|z_h)$ in jet12_double MC at 1.5 mrad (with $z_{\rm mb}$ marginalized), with the 1.5 mrad data reco z_r shape overlaid. The kernel's $\sim$34 cm RMS exceeds the $\sim$22 cm data peak, which is the geometric lower bound on truth-only reweighting.

D Files and code

- `anatreemaker/source/CaloAna24.{h,cc}` — writes the `vertexz_truth` (z_h, primary truth vertex) and `vertexz_truth_second` ($z_{\rm mb}$, secondary truth vertex; read by C++ as `vertexz_truth_mb`) branches that the iterative method reads for single- and double-interaction MC respectively.
- `efficiencytool/truth_vertex_reweight/` — the standalone Python tool for the iterative method:
  - `fit_core.py` — cached sample loader, the `mixed_mc_histogram` forward model (Eq. 2), Poisson χ^2, and helpers for fitting $\sigma_{\rm gen}$ from MC truth.
  - `fit_truth_vertex_reweight.py` — the driver (stage iterative = Section 3), producing `reweight.root` and `fit_result.yaml`.
  - `plot_closure.py`, `plot_comparisons.py` — closure and cross-period diagnostic plots.
  - `config.yaml` — sample paths, run ranges, fit parameters.
- `efficiencytool/TruthVertexRecoCheck.C` — Appendix A supporting macro.
- Outputs: `efficiencytool/truth_vertex_reweight/output/{1p5mrad,0mrad}/reweight.root`, `fit_result.yaml`, and the cached event arrays in `output/cache/`.